EXPERIMENT 3

NETWORK PROTOCOL ANALYSIS USING WIRESHARK

Date: 31/12/2025

AIM

To familiarise the working of packet sniffing and analysis tool – Wireshark.

THEORY

Wireshark is a highly versatile and widely used open-source network packet analyzer. It lets users capture, inspect, and analyze network traffic in real time or from saved files. Wireshark is a powerful tool for network troubleshooting, monitoring, and security analysis. It provides detailed information about individual network packets, including their source, destination, protocol, payload, and timing, making it an invaluable resource for diagnosing network issues, identifying anomalies, and enhancing network security. Wireshark is available for multiple platforms, including Linux, Windows, and macOS, and is commonly employed by network administrators, security professionals, and developers to gain insights into network behavior and resolve connectivity and security challenges.

Wireshark can be used in the following ways:

- It is used by network security engineers to examine security problems.
- It allows the users to watch all the traffic being passed over the network.
- It is used by network engineers to troubleshoot network issues.
- It also helps to troubleshoot latency issues and malicious activities on your network.
- It can also analyze dropped packets.
- It helps us to know how all the devices like laptop, mobile phones, desktop, switch, routers, etc., communicate in a local network or the rest of the world. **Features of Wireshark:**
- **Multi-platform software:** Wireshark runs on multiple operating systems, including Linux, Windows, and macOS.

- **Three-pane packet browser:** It uses a standard interface with three panes to present packet details clearly.
- **Deep protocol inspection:** Wireshark can deeply inspect hundreds of network protocols.
- **Live network analysis:** It allows real-time data capture from various network types, such as Ethernet and loopback
- **Sorting and filtering options:** Users can easily filter and sort captured data for focused analysis.
- **VoIP analysis:** Wireshark includes specialized tools for analyzing VoIP traffic.
- **USB traffic capture:** It can capture raw USB traffic, providing broader analysis beyond traditional networks.
- **Customizable settings:** Filters, timers, and other settings allow precise control over what data is captured and displayed.
- **File format support:** It supports well-documented capture file formats such as PcapNg and Libpcap, which are used to store captured data.
- **Top software for network analysis:** Wireshark is highly regarded and widely used for tasks such as tracing unauthorized traffic, analysing firewall settings, and troubleshooting network issues

ALGORITHM

1. Double-click on wireshark to open it.
2. We will get the traffic traversing through the current interface.
3. Select the wi-fi option
4. Click Capture from the menu bar and click start**.** Then it shows the traffic with the packet content written in hexadecimal or ASCII format. And the information above the packet content, is the details of the packet header. It will continue listening to all the data packets, and you will get a lot of data. If you want to see a particular data, then you can click on the red button. The traffic will be stationary, and you can note the parameters like time, source, destination, the protocol being used, length, and the Info. To view in-depth details, you can click on that particular address; a lot of the information will be displayed below that. There will be detailed information on HTTP packets, TCP packets, etc. The red button is shown below:

The screen/interface of the Wireshark is divided into five parts:

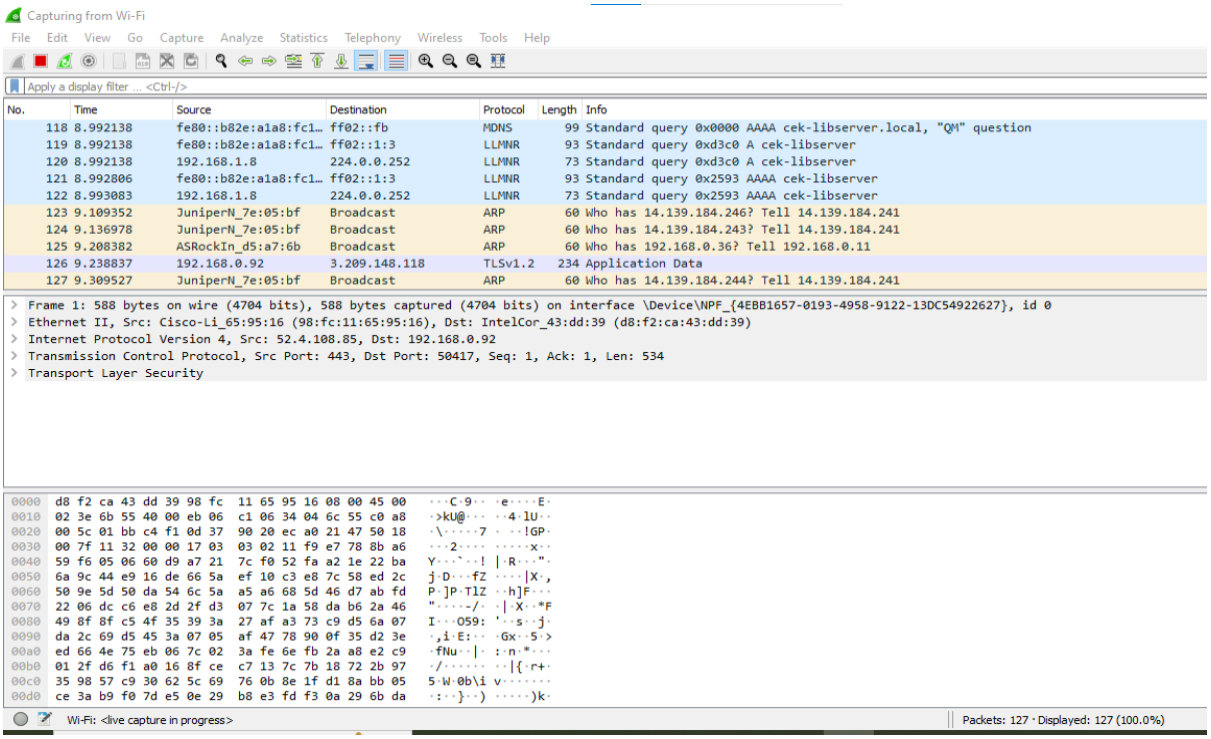
- First part contains a menu bar and the options displayed below it. This part is at the top of the window. File and the capture menus options are commonly used in Wireshark. The capture menu allows to start the capturing process. And the File menu is used to open and save a capture file.
- The second part is the packet listing window. It determines the packet flow or the captured packets in the traffic. It includes the packet number, time, source, destination, protocol, length, and info. We can sort the packet list by clicking on the column name.
- Next comes the packet header- detailed window. It contains detailed information about the components of the packets. The protocol info can also be expanded or minimized according to the information required.
- The bottom window called the packet contents window, which displays the content in ASCII and hexadecimal format.
- At last, is the filter field which is at the top of the display. The captured packets on the screen can be filtered based on any component according to your requirements. For example, if we want to see only the packets with the HTTP protocol, we can apply filters to that option. All the packets with HTTP as the protocol will only be displayed on the screen.

5. There is a filter block below the menu bar, from where a large amount of data can be filtered. For example, if we apply a filter for HTTP, only the interfaces with the HTTP will be listed. We can apply filters for TCP,UDP ,ARP,ICMP

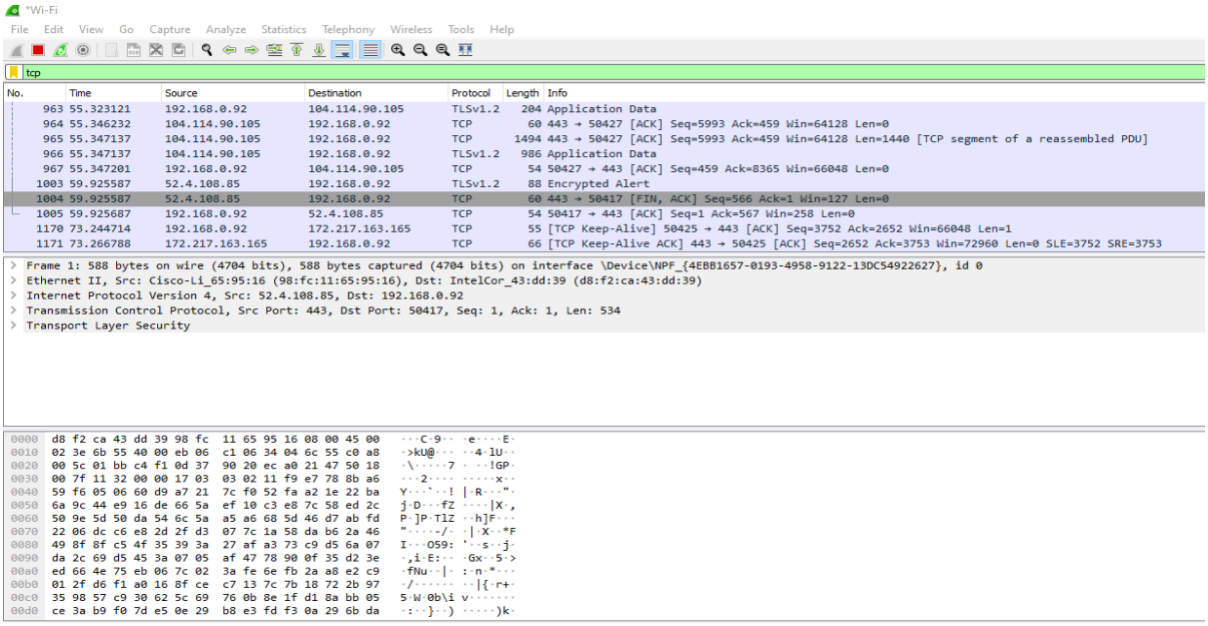
6. Steps for the permanent colorization are: click on the 'View' option on the menu bar and select 'Coloring Rules'**.** Select the option 'View' and then choose 'Colorize Packet List' which is used to toggle the color on and off. Whenever we type any commands in the filter command box, it turns green if your command is correct. It turns red if it is incorrect or the Wireshark does not recognize your command

SCREENSHOTS

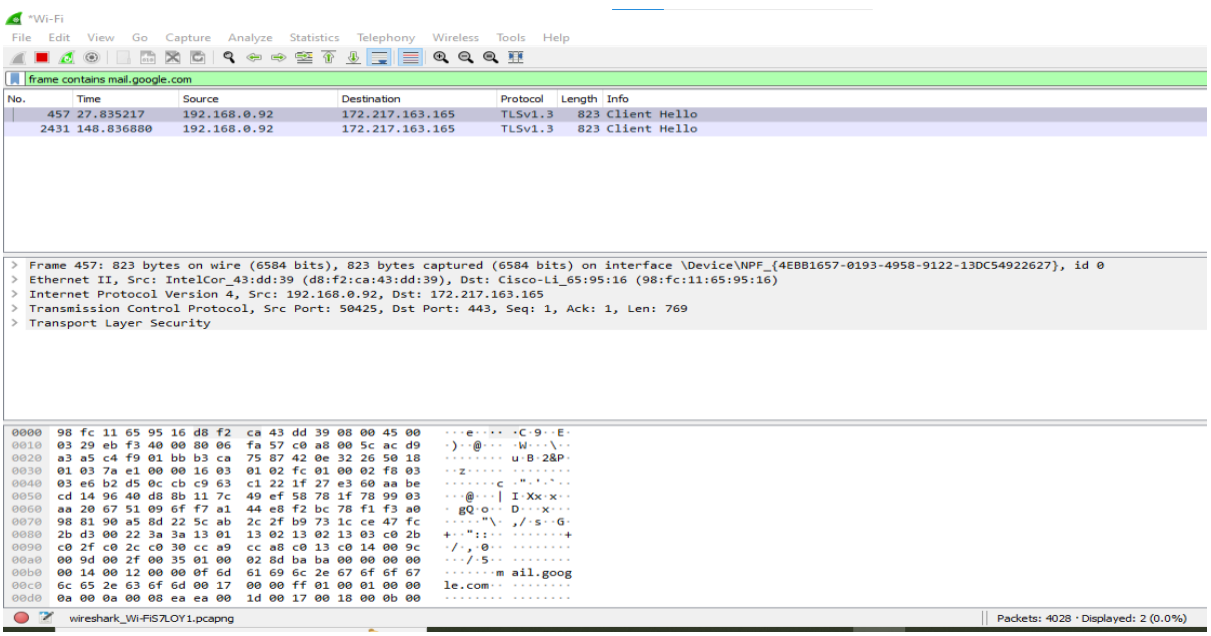
1. Open Wireshark tool



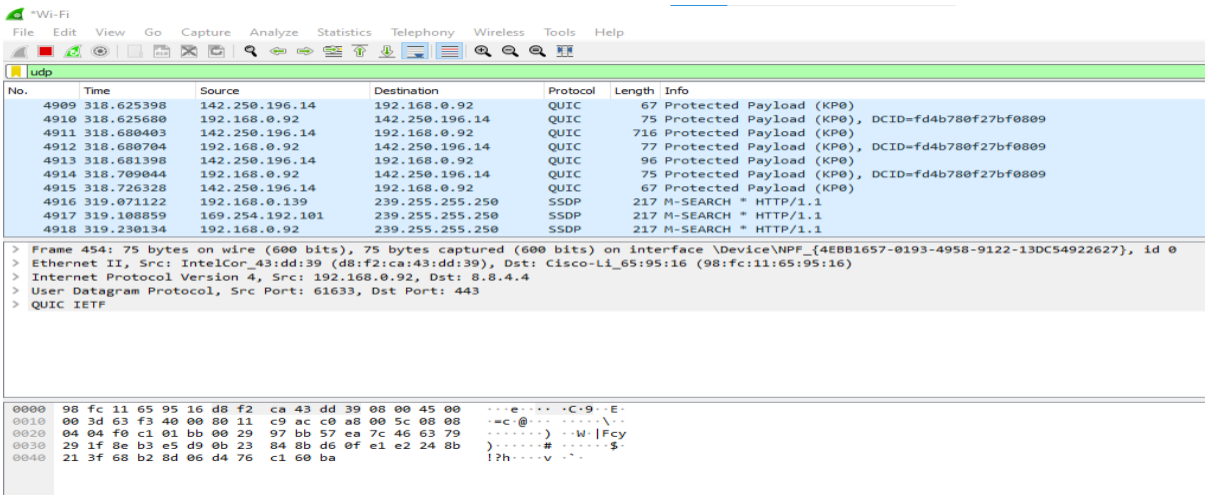
1. Capturing TCP packets



2. Capturing packets while checking gmail



3. UDP stream Analysing.



RESULT

Familiarised Wireshark and captured different network packet traffic.